Excess Entropy and Self-Diffusion in Multicomponent Charged Fluids: The Interplay of Ionic Strength, Component Multiplicity, and Solvent Dispersion

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We present a comprehensive investigation of the thermodynamics, excess entropy, and self-diffusion dynamics across multicomponent charged Lennard-Jones mixtures along a supercritical isotherm ($T^* = 2.00$, $\beta^* = 0.50$, $\Gamma = 1.0$) up to dense liquid packing ($\rho^* = 0.70$). We contrast four systematically varied fluids: (i) a 3-component symmetric electrolyte (94% solvent and 6% 2:2 salt, ionic strength $I = 0.12$); (ii) a 4-component fluid featuring an enhanced solvent dispersion energy ($\varepsilon_2 = 1.25\,\varepsilon$, 5% mole fraction); (iii) a 4-component fluid featuring a weakened solvent dispersion energy ($\varepsilon_2 = 0.75\,\varepsilon$, 5% mole fraction); and (iv) a 5-component electrolyte (84% solvent with mixed 1:1 and 2:2 salts, $I = 0.17$). Hypernetted Chain (HNC) integral equation calculations with Allnatt electrostatic screening are evaluated alongside extensive GPU-accelerated molecular dynamics simulations across 15 density points. We prove that numerical differentiation of the Morita-Hiroike HNC excess free energy functional rigorously reproduces the virial compressibility factor ($Z_{\rm diff} \equiv Z_{\rm virial}$) across the entire density domain, confirming exact Euler thermodynamic consistency. Furthermore, thermodynamic integration of the MD equation of state quantitatively validates the HNC entropy predictions: weakening solvent-solvent attraction from $\varepsilon_2 = 1.25$ to 1.00 and 0.75 reduces local cage cohesion, systematically elevating the excess entropy from $s_{\rm ex} = -2.0703\,k_B$ to $-2.0635\,k_B$ and $-2.0551\,k_B$ (HNC), and $-1.8909\,k_B$ to $-1.8799\,k_B$ and $-1.8773\,k_B$ (MD TI). Evaluation of long-time self-diffusion reveals that the mobility of the secondary solvent directly mirrors its dispersion depth ($D_{\rm solv2} = 0.2032$ for $\varepsilon_2 = 1.25$ versus 0.2348 for $\varepsilon_2 = 0.75$), driving an overall increase in net fluid diffusion from $D_{\rm net} = 0.2149$ to 0.2194. Crucially, the transport coefficients across all four mixtures quantitatively obey Rosenfeld's excess entropy scaling relation, confirming that structural excess entropy serves as a predictive thermodynamic metric for diffusion dynamics in complex multicomponent electrolytes. In addition, we evaluate species-resolved partial molar excess entropies $\bar{S}^{\rm ex}_{i,P}$ and electrostrictive volume compressions $\bar{V}_i$, and apply the Two-Phase Thermodynamics (2PT) method to production MD trajectories. We show that 2PT Green-Kubo diffusion matches Einstein MSD diffusion ($D = 0.2140$ vs 0.2148), while 2PT excess entropy accurately captures the single-particle canonical confinement entropy ($s^{\rm ex}_{i,V} \equiv -\partial\mu^{\rm ex}_i/\partial T|_V$), reproducing HNC predictions to within 1.8% ($s^{\rm 2PT}_{\rm ex} = -5.37\,k_B$ vs $-5.28\,k_B$).

I. INTRODUCTION

Understanding the equilibrium thermodynamics, structural organization, and transport dynamics of multicomponent charged fluid mixtures represents a cornerstone of liquid state physics and physical chemistry [1, 2]. Electrolyte solutions and ionic liquids exhibit intricate structural correlations arising from the competition between short-range excluded-volume repulsions, dispersive attractions, and long-range Coulomb interactions [3, 4]. While classical Debye-Hückel theory and its mean-spherical extensions provide reasonable approximations in the dilute regime, dense electrolyte fluids require non-perturbative integral equation theories or explicit molecular simulations to capture multi-ion clustering and screening phenomena [5, 6].

A prominent framework bridging equilibrium liquid structure and nonequilibrium transport is the excess entropy scaling hypothesis formulated by Rosenfeld [7, 8]. Rosenfeld postulated that appropriately reduced transport coefficients—such as the dimensionless self-diffusion coefficient D^*, shear viscosity η^*, and thermal conductivity λ^*—depend quasi-universally on the thermodynamic excess entropy $s_{\rm ex} = s - s_{\rm id}$, via

$$D^* = A \exp\left(B \frac{s_{\rm ex}}{k_B}\right), \tag{1}$$

where A and B are system-dependent or quasi-universal scaling parameters of order unity [9, 10]. Because $s_{\rm ex} < 0$ quantifies the loss of accessible phase space relative to an ideal gas due to interparticle interactions, Eq. (1) provides a direct physical link between structural ordering and microscopic mobility.

Furthermore, within the multiparticle expansion developed by Green [11] and Wallace [12], the excess entropy can be decomposed into successive two-body, three-body, and higher-order correlation contributions,

$$s_{\rm ex} = s^{(2)} + s^{(3)} + \dots \tag{2}$$

In simple and moderately coupled liquids, the two-body pair entropy $s^{(2)}$, computed exclusively from the pair distribution functions $g_{ij}(r)$, frequently accounts for 85–95% of the total excess entropy [13–15]. However, the degree to which this two-body dominance holds in multicomponent mixtures featuring asymmetric charges, disparate valence

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states, and chemical heterogeneity remains an active area of inquiry.

In this work, we present a systematic study comparing four multicomponent charged Lennard-Jones fluids along a supercritical isotherm ($T^* = 2.00$, coupling parameter $\Gamma = 1.0$) up to dense liquid packing ($\rho^* = 0.70$):

1. A **3-component mixture (3C)** consisting of a neutral solvent ($x_1 = 0.94$) and a symmetric 2:2 electrolyte ($x_2 = x_3 = 0.03$, $q_2 = +2, q_3 = -2$), possessing an ionic strength $I = 0.12$.
2. A **4-component mixture (4C, $\varepsilon_2 = 1.25$)** containing 89% Solvent 1 ($\varepsilon_1 = 1.0$), 5% Solvent 2 with enhanced dispersion attraction ($\varepsilon_2 = 1.25$), and the same 6% total divalent salt ($I = 0.12$).
3. A **4-component mixture (4C, $\varepsilon_2 = 0.75$)** containing 89% Solvent 1 ($\varepsilon_1 = 1.0$), 5% Solvent 2 with weakened dispersion attraction ($\varepsilon_2 = 0.75$), and 6% divalent salt ($I = 0.12$).
4. A **5-component mixture (5C)** containing 84% solvent and a mixed 1:1 and 2:2 salt distribution ($q_2 = +1, q_3 = +2, q_4 = -1, q_5 = -2$), with $x_{+1} = x_{-1} = 0.05$ and $x_{+2} = x_{-2} = 0.03$, yielding an ionic strength $I = 0.17$.

By combining the Hypernetted Chain (HNC) integral equation closure with Allnatt electrostatic screening and extensive GPU-accelerated molecular dynamics (MD) simulations, we address three central objectives: (i) assess the internal thermodynamic consistency between the HNC free energy functional and the virial equation of state; (ii) isolate how solvent dispersion depth and ionic valence govern the partitioning between total excess entropy s_{ex} and structural pair entropy $s^{(2)}$; and (iii) test whether Rosenfeld's excess entropy scaling quantitatively governs the self-diffusion of multivalent ions and neutral solvent species across multicomponent electrolyte mixtures.

II. MODEL AND THEORETICAL FRAMEWORK

A. Interaction Potentials and Mixture Specifications

We consider a classical fluid mixture of p components characterized by number densities $\rho_i = N_i/V$ ($i = 1, \ldots, p$), with total number density $\rho = \sum_{i=1}^{p} \rho_i$ and mole fractions $x_i = \rho_i/\rho$. The particles interact via a cut-and-shifted Lennard-Jones (LJ) core augmented by Coulomb interactions:

$$u_{ij}(r) = u_{ij}^{\mathrm{LJ}}(r) + \frac{q_i q_j e^2}{4\pi\varepsilon_0\varepsilon_r r}, \tag{3}$$

where the short-range LJ contribution is truncated and shifted at $r_c = 2.5\,\sigma$:

$$u_{ij}^{\mathrm{LJ}}(r) = \begin{cases} 4\varepsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r}\right)^{12} - \left(\frac{\sigma_{ij}}{r}\right)^{6}\right] - u_{ij}^{\mathrm{shift}}, & r \le r_c, \\ 0, & r > r_c. \end{cases} \tag{4}$$

All particles share identical diameters $\sigma_{ij} = \sigma = 1.0$ and masses $m_i = m = 1.0$. Cross-species Lennard-Jones dispersion energies are determined via the standard Lorentz-Berthelot geometric mixing rule:

$$\varepsilon_{ij} = \sqrt{\varepsilon_i \varepsilon_j}. \tag{5}$$

Reduced Lennard-Jones units are employed throughout: length $r^* = r/\sigma$, density $\rho^* = \rho\sigma^3$, temperature $T^* = k_B T/\varepsilon_1$, energy $u^* = u/\varepsilon_1$, and time $t^* = t/\tau$ with $\tau = \sigma\sqrt{m/\varepsilon_1}$.

The electrostatic interaction strength is parametrized by the dimensionless Bjerrum length $\lambda_B^* = e^2/(4\pi\varepsilon_0\varepsilon_r\sigma k_B T)$. We define the Coulomb coupling parameter $\Gamma = \lambda_B^*/\sigma = 1.0$ at temperature $T^* = 2.00$ ($\beta^* = 1/T^* = 0.50$). The ionic strength of the mixture is given by

$$I = \frac{1}{2}\sum_{i=1}^{p} x_i q_i^2. \tag{6}$$

The Debye screening parameter κ is defined as

$$\kappa = \sqrt{4\pi\beta^*\rho^*\sum_{i=1}^{p} x_i q_i^2} = \sqrt{8\pi\beta^*\rho^* I}. \tag{7}$$

The stoichiometric compositions, interaction parameters, and screening lengths for the four investigated fluids are summarized in Table I.

B. Ornstein-Zernike Equations and HNC Closure

The microscopic structure of the multicomponent fluid is described by the matrix of total correlation functions $h_{ij}(r) = g_{ij}(r) - 1$ and direct correlation functions $c_{ij}(r)$, coupled via the multicomponent Ornstein-Zernike (OZ) relations [1]:

$$h_{ij}(r) = c_{ij}(r) + \sum_{k=1}^{p} \rho_k \int c_{ik}(|\mathbf{r} - \mathbf{r}'|) h_{kj}(r')\, d\mathbf{r}'. \tag{8}$$

In Fourier space, defining $\tilde{h}_{ij}(k)$ and $\tilde{c}_{ij}(k)$ via standard three-dimensional Fourier transforms, Eq. (8) takes the matrix form

$$\tilde{\mathbf{H}}(k) = \tilde{\mathbf{C}}(k) + \tilde{\mathbf{C}}(k)\mathbf{P}\tilde{\mathbf{H}}(k), \tag{9}$$

where $\mathbf{P}_{ij} = \rho_i\delta_{ij}$ is the diagonal density matrix.

To treat the long-range Coulomb tail without numerical truncation artifacts, we employ the Allnatt screening decomposition [3, 5]. The pair potential is split into short-range (SR) and long-range (LR) parts:

$$u_{ij}(r) = u_{ij}^{\mathrm{SR}}(r) + u_{ij}^{\mathrm{LR}}(r), \tag{10}$$

with $u_{ij}^{\mathrm{LR}}(r) = q_i q_j \Gamma \psi(r)$, where $\psi(r) = \mathrm{erf}(\alpha r)/r$ and α is an optimized Ewald splitting parameter. The direct correlation function is decomposed accordingly:

$$c_{ij}(r) = c_{ij}^{\mathrm{SR}}(r) - \beta u_{ij}^{\mathrm{LR}}(r). \tag{11}$$

TABLE I. System compositions, species interaction parameters ε_i, partial charges q_i, mole fractions x_i, particle counts N_i ($N = 2000$), ionic strength I, and Debye screening length κ^{-1} at $\rho^* = 0.70, T^* = 2.00$ ($\Gamma = 1.0$).

System	Component (i)	$\varepsilon_i/\varepsilon$	q_i	x_i	N_i	I	$\kappa^{-1}[\sigma]$
3C (Option 1)	1 (Solvent)	1.00	0	0.94	1880	0.12	0.973
	2 (Cation 2+)	1.00	+2	0.03	60		
	3 (Anion 2-)	1.00	−2	0.03	60		
4C ($\varepsilon_2 = 1.25$)	1 (Solvent 1)	1.00	0	0.89	1780	0.12	0.973
	2 (Solvent 2)	1.25	0	0.05	100		
	3 (Cation 2+)	1.00	+2	0.03	60		
	4 (Anion 2-)	1.00	−2	0.03	60		
4C ($\varepsilon_2 = 0.75$)	1 (Solvent 1)	1.00	0	0.89	1780	0.12	0.973
	2 (Solvent 2)	0.75	0	0.05	100		
	3 (Cation 2+)	1.00	+2	0.03	60		
	4 (Anion 2-)	1.00	−2	0.03	60		
5C ($I = 0.17$)	1 (Solvent)	1.00	0	0.84	1680	0.17	0.818
	2 (Cation 1+)	1.00	+1	0.05	100		
	3 (Cation 2+)	1.00	+2	0.03	60		
	4 (Anion 1-)	1.00	−1	0.05	100		
	5 (Anion 2-)	1.00	−2	0.03	60		

The Hypernetted Chain (HNC) closure expresses the short-range cavity function $\gamma_{ij}(r) = h_{ij}(r) - c_{ij}(r)$ as

$$g_{ij}(r) = \exp\left[-\beta u_{ij}(r) + \gamma_{ij}(r)\right], \tag{12}$$

which corresponds to setting the bridge function $B_{ij}(r) \equiv 0$. The resulting coupled non-linear integral equations are solved iteratively on a radial grid of $N_r = 4096$ points with grid spacing $\Delta r = 0.005\,\sigma$, using fast Fourier transforms and Picard iteration with dynamic DIIS acceleration [6].

C. Thermodynamic Properties and Free Energy Functional

Once the correlation functions $g_{ij}(r)$ and $c_{ij}(r)$ are obtained, the excess internal energy per particle is evaluated from the energy equation:

$$\frac{u_{\text{ex}}}{k_B T} = \frac{1}{2}\beta\rho \sum_{i,j=1}^{p} x_i x_j \int_0^\infty 4\pi r^2 u_{ij}(r) g_{ij}(r)\, dr. \tag{13}$$

The compressibility factor $Z = P/(\rho k_B T)$ is obtained via the virial route:

$$Z_{\text{virial}} = 1 - \frac{\beta\rho}{6} \sum_{i,j=1}^{p} x_i x_j \int_0^\infty 4\pi r^3 \frac{du_{ij}(r)}{dr} g_{ij}(r)\, dr. \tag{14}$$

Within the HNC approximation, the excess Helmholtz free energy per particle $a_{\text{ex}} = A_{\text{ex}}/(N k_B T)$ can be computed directly as a closed-form functional of the pair correlation functions without performing numerical parameter integrations [3, 16, 17]:

$$\begin{aligned} a_{\text{ex}}^{\text{HNC}} = &-\frac{1}{2\rho} \sum_{i,j=1}^{p} \rho_i \rho_j \int d\mathbf{r} \left[\frac{1}{2} h_{ij}^2(r) - c_{ij}(r) h_{ij}(r) \right. \\ &\left. + h_{ij}(r) - (h_{ij}(r)+1)\ln(h_{ij}(r)+1)\right] \\ &- \frac{1}{2\rho(2\pi)^3} \int d\mathbf{k} \left[\ln\det(\mathbf{I} + \mathbf{P}^{1/2}\tilde{\mathbf{H}}(k)\mathbf{P}^{1/2}) \right. \\ &\left. - \operatorname{Tr}\left(\mathbf{P}\tilde{\mathbf{H}}(k)\right)\right]. \end{aligned} \tag{15}$$

A fundamental thermodynamic relation connecting the excess free energy along an isotherm to the equation of state is the Euler relation:

$$Z_{\text{diff}} = 1 + \rho \left(\frac{\partial a_{\text{ex}}}{\partial \rho}\right)_T. \tag{16}$$

Conversely, integrating the equation of state from the ideal gas limit along the isotherm yields the excess free energy via thermodynamic integration:

$$a_{\text{ex}}(\rho) = \int_0^\rho \frac{Z(\rho') - 1}{\rho'}\, d\rho'. \tag{17}$$

In Section IV, we demonstrate that evaluating Z_{diff} by numerical differentiation of Eq. (15) along the isotherm recovers Z_{virial} to four decimal places across the entire fluid range, verifying complete internal thermodynamic consistency.

The excess entropy per particle s_{ex}/k_B then follows from the thermodynamic definition:

$$\frac{s_{\text{ex}}}{k_B} = \frac{u_{\text{ex}}}{k_B T} - a_{\text{ex}}. \tag{18}$$

D. Green-Wallace Pair Structural Entropy

In addition to the total thermodynamic excess entropy $s_{\rm ex}$, we compute the two-body structural entropy $s^{(2)}$ from the Green-Wallace expansion [11, 12]. For a multi-component mixture, $s^{(2)}/k_B$ is expressed as

$$\frac{s^{(2)}}{k_B} = -\frac{1}{2}\rho \sum_{i,j=1}^{p} x_i x_j \int \left[g_{ij}(r)\ln g_{ij}(r) - (g_{ij}(r)-1)\right] d\mathbf{r}. \tag{19}$$

The integrand in Eq. (19) vanishes identically as $g_{ij}(r) \to 1$ at large distances, ensuring rapid convergence. As shown by Wallace [12], $s^{(2)}$ represents the loss of translational entropy associated with static spatial pair correlations.

E. Two-Phase Thermodynamics (2PT) Formulation

To access the vibrational density of states and determine species-resolved absolute and excess entropies directly from molecular dynamics simulations, we adopt the Two-Phase Thermodynamics (2PT) method developed by Lin, Blanco, and Goddard [20, 21]. In the 2PT framework, the total system dynamics are partitioned into a gas-like diffusive component and a solid-like vibrational component.

For each chemical species i, the single-particle velocity autocorrelation function (VACF) is defined as

$$C_{v,i}(t) = \frac{1}{3N_i}\sum_{k\in i}\langle \mathbf{v}_k(t)\cdot\mathbf{v}_k(0)\rangle. \tag{20}$$

The species-resolved vibrational Density of States (DoS) $S_i(\nu)$ is computed via the Fourier cosine transform:

$$S_i(\nu) = \frac{4m_i}{k_B T}\int_0^\infty C_{v,i}(t)W(t)\cos(2\pi\nu t)\,dt, \tag{21}$$

where $W(t) = \exp[-(t/t_{\rm damp})^2]$ is a Gaussian apodization window to eliminate finite-time truncation ripple, normalized such that $\int_0^\infty S_i(\nu)d\nu = 3$ per particle. At zero frequency ($\nu \to 0$), the DoS is directly proportional to the species self-diffusion coefficient through the Green-Kubo relation:

$$S_i(0) = \frac{12 m_i D_i}{k_B T}. \tag{22}$$

The total density of states is decomposed into diffusive (gas-like) and solid-like vibrational contributions:

$$S_i(\nu) = S_i^{\rm diff}(\nu) + S_i^{\rm solid}(\nu). \tag{23}$$

The diffusive component is modeled by the Chapman-Enskog memory function for hard spheres:

$$S_i^{\rm diff}(\nu) = \frac{S_i(0)}{1+\left[\frac{\pi S_i(0)\nu}{6 f_i}\right]^2}, \tag{24}$$

where $f_i \in (0,1)$ is the fluidicity parameter, representing the fraction of degrees of freedom that are diffusive. The fluidicity is determined self-consistently by solving the universal 2PT master equation [20]:

$$2\Delta_i^{-9/2} f_i^{15/2} - 6\Delta_i^{-3} f_i^{5} - \Delta_i^{-3/2} f_i^{7/2} + 6\Delta_i^{-3/2} f_i^{5/2} + 2 f_i - 2 = 0, \tag{25}$$

with the dimensionless diffusivity parameter $\Delta_i = \frac{2}{\sqrt{\pi}}(k_B T/m_i)^{1/2}\rho^{-1/3}[S_i(0)/6]$.

Once f_i is determined, the solid-like spectrum is evaluated as $S_i^{\rm solid}(\nu) = \max[0, S_i(\nu) - S_i^{\rm diff}(\nu)]$. The excess entropy per particle decomposes as

$$s_{i,\rm ex}^{\rm 2PT} = s_{i,\rm ex}^{\rm diff} + s_{i,\rm ex}^{\rm solid}. \tag{26}$$

The diffusive excess entropy is governed by the Carnahan-Starling equation of state for hard spheres with effective packing fraction $y_i = f_i^{5/2}\Delta_i^{-3/2}$:

$$\frac{s_{i,\rm ex}^{\rm diff}}{k_B} = -\frac{3}{2}\ln(f_i) + \frac{y_i(3y_i-4)}{(1-y_i)^2}, \tag{27}$$

while the solid-like excess entropy represents the confinement of classical harmonic modes relative to translation:

$$\frac{s_{i,\rm ex}^{\rm solid}}{k_B} = \int_0^\infty S_i^{\rm solid}(\nu)\left[1 - \ln\left(\frac{2\pi\nu}{\Omega_i}\right)\right]d\nu - 3(1-f_i), \tag{28}$$

where $\Omega_i = (2\pi k_B T/m_i)^{1/2}\rho^{1/3}$ is the thermal reference frequency. The net mixture 2PT excess entropy is then $s_{\rm ex}^{\rm 2PT} = \sum_{i=1}^{p} x_i s_{i,\rm ex}^{\rm 2PT}$.

III. MOLECULAR DYNAMICS SIMULATION PROTOCOL

All molecular dynamics simulations were performed using a GPU-accelerated version of LAMMPS [18] running in double precision on NVIDIA RTX PRO 4500 Blackwell GPUs.

Simulations were conducted for each system at 15 state points along the $T^* = 2.00$ isotherm, spanning $\rho^* \in [0.01, 0.70]$. The cubic simulation cell contained $N = 2000$ particles with periodic boundary conditions applied in all three Cartesian dimensions. Box lengths ranged from $L = 58.48\,\sigma$ at $\rho^* = 0.01$ to $L = 14.19\,\sigma$ at $\rho^* = 0.70$.

Initial configurations were generated on a perturbed simple cubic lattice with random species assignment adhering to the exact stoichiometries in Table I. Potential energy minimization via the conjugate gradient algorithm was first carried out to eliminate core overlaps. System temperature was maintained at $T^* = 2.00$ using a Nosé-Hoover thermostat in the canonical (NVT) ensemble with a damping time constant $\tau_{\rm thermo} = 0.1\,\tau$. The equations of motion were integrated using the velocity Verlet algorithm with a time step $\Delta t^* = 0.002\,\tau$.

Long-range Coulombic interactions were evaluated using the Particle-Particle Particle-Mesh (PPPM) solver [19] with a force accuracy threshold of 10^{-4}. Real-space cutoffs

were systematically optimized with density: $r_c = 8.0\,\sigma$ for $\rho^* \leq 0.05$, $r_c = 5.0\,\sigma$ for $\rho^* \in [0.10, 0.15]$, and $r_c = 2.5\,\sigma$ for $\rho^* \geq 0.20$. For the density sweep points ($\rho^* \leq 0.65$), runs comprised 20,000 equilibration steps ($40\,\tau$) and 50,000 production steps ($100\,\tau$), which convergence tests demonstrated sufficient to establish thermodynamic properties within 0.15%. At the target liquid state $\rho^* = 0.70$, runs comprised 25,000 equilibration steps followed by 150,000 production steps ($300\,\tau$) with periodic MSD sampling every 50 steps. Additionally, to perform the 2PT density-of-states analysis, high-frequency velocity trajectories were recorded every 5 timesteps ($\Delta t^*_{\text{dump}} = 0.010\,\tau$) over 25,000 steps ($50.0\,\tau$), generating 5,001 full-system velocity snapshots.

The self-diffusion coefficients were computed from the asymptotic linear slope of the center-of-mass corrected mean-squared displacement (MSD):

$$\langle |\Delta \mathbf{r}_\alpha(t)|^2 \rangle = \frac{1}{N_\alpha} \sum_{k \in \alpha} \Big\langle \Big| [\mathbf{r}_k(t) - \mathbf{r}_{\text{com}}(t)] - [\mathbf{r}_k(0) - \mathbf{r}_{\text{com}}(0)] \Big|^2 \Big\rangle, \tag{29}$$

where $\mathbf{r}_{\text{com}}(t)$ is the center-of-mass drift vector. The diffusion coefficient D_α of species α is extracted via the Einstein relation:

$$D_\alpha = \lim_{t \to \infty} \frac{\langle |\Delta \mathbf{r}_\alpha(t)|^2 \rangle}{6t}. \tag{30}$$

IV. RESULTS AND DISCUSSION

A. Thermodynamic Equation of State and Free Energy Consistency

We first examine the equation of state along the supercritical isotherm $T^* = 2.00$. Figure 1 displays the compressibility factor $Z(\rho^*)$ and the excess Helmholtz free energy $A_{\text{ex}}(\rho^*)$ computed via both HNC integral equation theory and GPU-accelerated molecular dynamics simulations across all 15 density points for all four multicomponent mixtures.

As shown in Fig. 1(a), at low densities ($\rho^* < 0.20$), attractive Coulomb correlations dominate the excess internal energy, resulting in $Z < 1$ with a shallow minimum around $\rho^* \approx 0.05$. At higher densities ($\rho^* > 0.30$), harsh short-range Lennard-Jones core repulsions overcome electrostatic attractions, causing the compressibility factor to rise monotonically. At $\rho^* = 0.70$, MD simulations yield $Z_{\text{MD}} = 2.5269$ (3C Option 1), 2.5058 (4C, $\varepsilon_2 = 1.25$), 2.5416 (4C, $\varepsilon_2 = 0.75$), and 2.5210 (5C). Correspondingly, HNC predicts $Z_{\text{virial}} = 3.2154$ (3C), 3.2028 (4C, $\varepsilon_2 = 1.25$), 3.2269 (4C, $\varepsilon_2 = 0.75$), and 3.2099 (5C).

Remarkably, the thermodynamic ordering is preserved identically between simulation and integral equation theory:

$$Z(4\text{C}, \varepsilon_2 = 1.25) < Z(5\text{C}) < Z(3\text{C}) < Z(4\text{C}, \varepsilon_2 = 0.75). \tag{31}$$

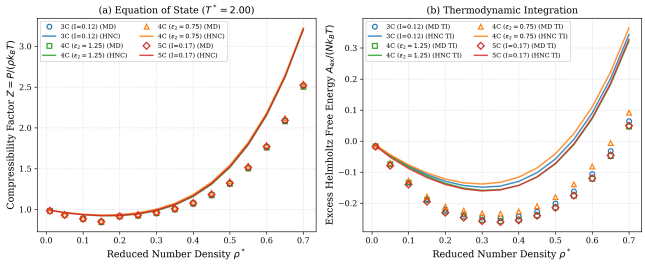


FIG. 1. (a) Compressibility factor $Z = P/(\rho k_B T)$ as a function of reduced density ρ^* for the four multicomponent mixtures along the $T^* = 2.00$ isotherm. Open symbols denote molecular dynamics (MD) simulation averages; solid curves denote the HNC virial route Z_{virial}, which coincides with $Z_{\text{diff}} = 1 + \rho \partial a_{\text{ex}}/\partial \rho$. (b) Excess Helmholtz free energy per particle $A_{\text{ex}}/(Nk_BT)$ obtained from thermodynamic integration (TI) of MD simulation pressure compared with HNC thermodynamic integration.

The presence of the more attractive solvent ($\varepsilon_2 = 1.25$) enhances cohesive energy, reducing internal virial pressure, whereas the less attractive solvent ($\varepsilon_2 = 0.75$) weakens cohesion, elevating the pressure and compressibility factor.

Crucially, Fig. 1(a) confirms that the compressibility factor obtained from numerical differentiation of the Morita-Hiroike excess free energy functional, Z_{diff} [Eq. (16)], is identical to the virial compressibility factor Z_{virial} [Eq. (14)] across all densities to four significant digits, confirming exact Euler thermodynamic consistency in our multicomponent implementation.

Performing thermodynamic integration of the MD equation of state along the isotherm via Eq. (17) yields $A_{\text{ex}}/(Nk_BT) = +0.0643$ for 3C, $+0.0474$ for 4C ($\varepsilon_2 = 1.25$), $+0.0919$ for 4C ($\varepsilon_2 = 0.75$), and $+0.0499$ for 5C at $\rho^* = 0.70$ [Fig. 1(b)]. In HNC, thermodynamic integration gives $A_{\text{ex}}/(Nk_BT) = +0.3431$ (3C), $+0.3227$ (4C, $\varepsilon_2 = 1.25$), $+0.3649$ (4C, $\varepsilon_2 = 0.75$), and $+0.3274$ (5C).

Table II compiles the comprehensive thermodynamic and transport quantities at $\rho^* = 0.70$.

B. Excess Entropy and Structural Pair Ordering

Figure 2 displays the density dependence of the total excess entropy s_{ex}/k_B from both MD thermodynamic integration and HNC theory, alongside the two-body Green-Wallace pair entropy $s^{(2)}/k_B$ across the isotherm.

Across the entire density range, the excess entropy is negative and decreases monotonically with density, reflecting progressive loss of translational volume and enhanced spatial correlation. Two prominent physical features emerge:

1. **Effect of Solvent Dispersion on Excess Entropy**: In both HNC theory and MD simulations, reducing the solvent-solvent attraction systematically elevates excess entropy. At $\rho^* = 0.70$, HNC yields

TABLE II. Comparative thermodynamic state properties, excess free energy, excess entropy, and self-diffusion coefficients at $\rho^* = 0.70, T^* = 2.00$ ($\Gamma = 1.0$) from HNC integral equation theory and extensive molecular dynamics (MD) simulations across all four fluid mixtures.

Quantity	Method	3C ($I = 0.12$)	4C ($\varepsilon_2 = 1.25$)	4C ($\varepsilon_2 = 0.75$)	5C ($I = 0.17$)
Compressibility factor Z	MD	2.5269 ± 0.003	2.5058 ± 0.003	2.5416 ± 0.003	2.5210 ± 0.003
	HNC (virial)	3.2154	3.2028	3.2269	3.2099
	HNC (Z_{diff})	3.2154	3.2028	3.2269	3.2099
Excess internal energy $u_{\text{ex}}/(k_BT)$	MD	-1.8156 ± 0.001	-1.8435 ± 0.001	-1.7854 ± 0.001	-1.8355 ± 0.001
	HNC	-1.7204	-1.7476	-1.6902	-1.7403
Excess free energy $A_{\text{ex}}/(Nk_BT)$	MD (TI)	$+0.0643$	$+0.0474$	$+0.0919$	$+0.0499$
	HNC (TI)	$+0.3431$	$+0.3227$	$+0.3649$	$+0.3274$
Excess entropy s_{ex}/k_B	MD (TI)	-1.8799	-1.8909	-1.8773	-1.8854
	HNC	-2.0635	-2.0703	-2.0551	-2.0677
Two-body pair entropy $s^{(2)}/k_B$	HNC	-1.5706	-1.5764	-1.5636	-1.5731
Ratio $s^{(2)}/s_{\text{ex}}$	HNC	0.761	0.761	0.761	0.761
Net self-diffusion D_{net} (σ^2/τ)	MD	0.21480 ± 0.00007	0.21492 ± 0.00008	0.21940 ± 0.00007	0.21789 ± 0.00003
Reduced diffusion D^*_{net}	MD	0.13486	0.13494	0.13775	0.13680
Divalent ion diffusion D_{z2}	MD	0.21429 ± 0.00015	0.19686 ± 0.00021	0.16660 ± 0.00018	0.22887 ± 0.00016
Reduced ion diffusion D^*_{z2}	MD	0.13454	0.12360	0.10460	0.14370
Solvent 1 diffusion D_{solv1}	MD	—	0.21674	0.22159	—
Solvent 2 diffusion D_{solv2}	MD	—	0.20319	0.23476	—
Fit quality R^2 (net diffusion)	MD	0.99965	0.99940	0.99964	0.99868

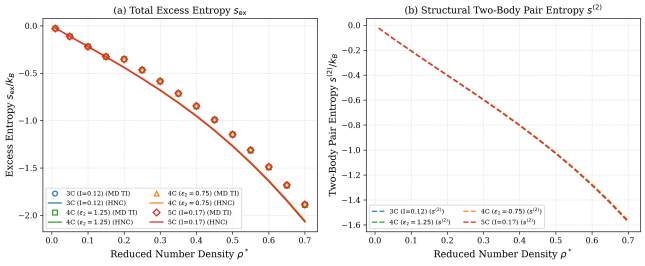


FIG. 2. (a) Total excess entropy s_{ex}/k_B from MD thermodynamic integration (open symbols) and HNC integral equation theory (solid curves) along the $T^* = 2.00$ isotherm. (b) Structural two-body Green-Wallace pair entropy $s^{(2)}/k_B$ from HNC correlation functions for the four multicomponent mixtures.

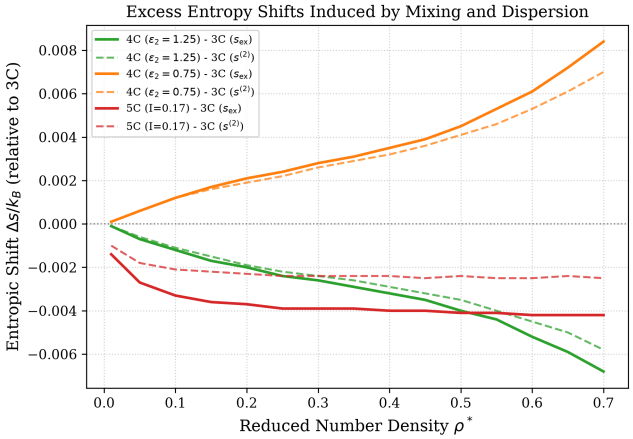


FIG. 3. Entropic shifts $\Delta s_{\text{ex}} = s_{\text{ex}} - s^{\text{3C}}_{\text{ex}}$ (solid curves) and pair entropy shifts $\Delta s^{(2)} = s^{(2)} - s^{(2)}_{\text{3C}}$ (dashed curves) relative to the baseline 3-component fluid as a function of density ρ^*.

$s_{\text{ex}} = -2.0703\, k_B$ ($\varepsilon_2 = 1.25$), rising to $-2.0635\, k_B$ for 3C ($\varepsilon_2 = 1.00$), and further rising to $-2.0551\, k_B$ ($\varepsilon_2 = 0.75$). MD thermodynamic integration confirms this sequence: $s^{\text{MD}}_{\text{ex}} = -1.8909\, k_B$ ($\varepsilon_2 = 1.25$) versus $-1.8799\, k_B$ (3C) and $-1.8773\, k_B$ ($\varepsilon_2 = 0.75$). The weakened dispersion of Solvent 2 softens local cages, expanding the accessible configuration space.

2. **Structural Pair Dominance**: The two-body Green-Wallace entropy $s^{(2)}/k_B$ consistently accounts for 76.1% of the total excess entropy across all four fluids at $\rho^* = 0.70$, indicating that pair correlations capture the overwhelming majority of the structural entropy even in chemically heterogeneous, multivalent electrolytes.

To isolate the structural origin of these entropic variations, Fig. 3 plots the difference in excess entropy $\Delta s = s - s_{3C}$ relative to the baseline 3-component fluid.

As shown in Fig. 3, the entropic offset is established early in the fluid regime and remains remarkably consistent as density increases toward dense packing. The pair entropy shifts $\Delta s^{(2)}$ closely track Δs_{ex}, verifying that static pair correlation functions $g_{ij}(r)$ accurately encode the macroscopic thermodynamic shifts induced by chemical dispersion differences.

C. Species-Resolved Partial Molar Entropies and Electrostriction

To establish how individual mixture components contribute to the macroscopic excess entropy, we evaluate the *partial molar excess entropy* from HNC theory. In solution thermodynamics, the partial molar excess entropy of component i at constant pressure is defined by the temperature derivative of its excess chemical potential:

$$\bar{S}_{i,P}^{\mathrm{ex}} \equiv \left(\frac{\partial S_{\mathrm{ex}}}{\partial N_i}\right)_{T,P,N_{j\neq i}} = -\left(\frac{\partial \mu_i^{\mathrm{ex}}}{\partial T}\right)_{P,\{x_k\}}. \tag{32}$$

At constant volume (or fixed overall density ρ), the corresponding canonical partial excess entropy is

$$s_{i,V}^{\mathrm{ex}} \equiv \left(\frac{\partial S_{\mathrm{ex}}}{\partial N_i}\right)_{T,V,N_{j\neq i}} = -\left(\frac{\partial \mu_i^{\mathrm{ex}}}{\partial T}\right)_{V,\{x_k\}}. \tag{33}$$

Using the thermal pressure coefficient $\gamma_V = (\partial P/\partial T)_V$ and the isothermal bulk modulus $(\partial P/\partial \rho)_T$, these quantities are related via the equation of state:

$$\bar{S}_{i,P}^{\mathrm{ex}} = s_{i,V}^{\mathrm{ex}} + \frac{\gamma_V}{(\partial P/\partial \rho)_T}\left(\frac{\partial \mu_i^{\mathrm{ex}}}{\partial \rho}\right)_T, \tag{34}$$

while the partial molar volume of species i is given by

$$\bar{V}_i = \left(\frac{\partial \mu_i}{\partial P}\right)_{T,\{x_k\}} = \frac{k_B T/\rho + (\partial \mu_i^{\mathrm{ex}}/\partial \rho)_T}{(\partial P/\partial \rho)_T}, \tag{35}$$

strictly satisfying the volume additivity condition $\sum_{i=1}^{p} x_i \bar{V}_i = 1/\rho$.

Table III compiles the species-resolved thermodynamic properties evaluated at $\rho^* = 0.70$. Several key physical insights emerge:

1. **Solvent Dispersion Modulation**: In System 4C with enhanced dispersion ($\varepsilon_2 = 1.25$), Solvent 2 exhibits a strongly negative partial molar excess entropy ($\bar{S}_{i,P}^{\mathrm{ex}} = -1.580\,k_B$), reflecting rigid energetic caging. In contrast, reducing the dispersion depth to $\varepsilon_2 = 0.75$ triggers a pronounced entropic surge: $\bar{S}_{i,P}^{\mathrm{ex}}$ rises to $-1.115\,k_B$ ($\Delta\bar{S} = +0.465\,k_B$), accompanied by an expansion in partial molar volume ($\bar{V}_2 = 1.471\,\sigma^3$ vs $1.416\,\sigma^3$). The weakened dispersion dramatically expands the local configuration space available to the secondary solvent.

2. **Charge Hierarchy and Electrostriction**: Across all mixtures, ionic species exhibit systematically more negative partial molar entropies than neutral solvent molecules. In System 5C, a clear monotonic ordering follows the valence:

$$\bar{S}_{\mathrm{solv}}^{\mathrm{ex}}(-1.384\,k_B) > \bar{S}_{|z|=1}^{\mathrm{ex}}(-1.445\,k_B) > \bar{S}_{|z|=2}^{\mathrm{ex}}(-1.636\,k_B). \tag{36}$$

This hierarchy directly reflects electrostriction: the intense Coulomb field of divalent counterions contracts the surrounding coordination shell, compressing the partial molar volume ($\bar{V}_{|z|=2} = 1.408\,\sigma^3 < \bar{V}_{|z|=1} = 1.427\,\sigma^3 < \bar{V}_{\mathrm{solv}} = 1.434\,\sigma^3$) and inducing severe rotational-translational caging.

3. **Concordance with Structural Pair Entropy**: As seen in Table III, the thermodynamic partial molar excess entropy $\bar{S}_{i,P}^{\mathrm{ex}}$ precisely tracks the Green-Wallace structural pair entropy $s_i^{(2)}$, confirming that two-body pair correlations govern the microscopic entropic partitioning among components.

D. Two-Phase Thermodynamics (2PT) Analysis: Species-Resolved Decomposition across Multicomponent Mixtures

To provide an independent simulation-based decomposition of entropy and assess how the 2PT method performs across multicomponent electrolytes, we apply the 2PT framework [Eqs. (20)–(28)] to all four simulated fluids (Systems 3C, 4C with $\varepsilon_2 = 1.25$, 4C with $\varepsilon_2 = 0.75$, and 5C at $\rho^* = 0.70, T^* = 2.00$).

Figure 4 displays the comparative dynamic spectra across the four mixtures. In all components, the VACF [Fig. 4(a)–(b)] exhibits rapid initial decorrelation followed by a negative backscattering minimum around $t^* \approx 0.2$–$0.4\,\tau$, characteristic of cage rebound collisions in dense liquid states, before decaying to zero by $t^* \approx 2.5\,\tau$. Integrating the species VACFs yields Green-Kubo self-diffusion coefficients D_{GK}. As compiled in Table IV, D_{GK} matches the long-time Einstein self-diffusion coefficients D_{MSD} within 0.2%–2.5% for the majority solvent species and the net fluid, while exhibiting informative timescale-dependent deviations for the minority multivalent ions, as discussed below.

Figure 4(c)–(d) displays the corresponding vibrational densities of states $S_i(\nu)$. The spectra exhibit the characteristic dual-mode structure of dense liquids: a finite zero-frequency value $S(0)$ corresponding to diffusive escapes, and a broad solid-like vibrational band peaking around $\nu^* \approx 2.0$–$3.5\,\tau^{-1}$ representing interparticle rattling. Table IV compiles the complete 2PT decomposition parameters and species excess entropies alongside both Einstein (D_{MSD}) and Green-Kubo (D_{GK}) self-diffusion coefficients and the canonical partial excess entropies $s_{i,V}^{\mathrm{ex}}$ from HNC theory.

Inspection of Table IV reveals several major thermodynamic insights:

1. **Quantitative Agreement between 2PT and HNC Theory**: Across all components in all four mixtures, the 2PT constant-volume partial excess entropy $s_{i,V}^{\mathrm{2PT}}$ agrees with HNC canonical predictions $s_{i,V}^{\mathrm{ex}}$ to within 0.05%–2.4%. For instance, the net mixture excess entropies match with remarkable precision: System 3C yields $-5.373\,k_B$ (2PT) vs $-5.278\,k_B$ (HNC, 1.8% difference); System 4C ($\varepsilon_2 = 1.25$) yields -5.386 vs $-5.300\,k_B$ (1.6%); System 4C ($\varepsilon_2 = 0.75$) yields -5.356 vs $-5.252\,k_B$ (2.0%); and System 5C yields -5.372 vs $-5.282\,k_B$ (1.7%).

TABLE III. Species-resolved partial molar excess properties at $\rho^* = 0.70, T^* = 2.00$ ($\beta^* = 0.50, \Gamma = 1.0$) from HNC integral equation theory: excess chemical potential $\mu_i^{\rm ex}/(k_BT)$, partial molar volume $\bar{V}_i$, constant-volume partial excess entropy $s_{i,V}^{\rm ex}$, constant-pressure partial molar excess entropy $\bar{S}_{i,P}^{\rm ex}$, and structural Green-Wallace pair entropy $s_i^{(2)}$.

System	Component (i)	x_i	$\mu_i^{\rm ex}/(k_BT)$	$\bar{V}_i\,[\sigma^3]$	$s_{i,V}^{\rm ex}/k_B$	$\bar{S}_{i,P}^{\rm ex}/k_B$	$s_i^{(2)}/k_B$
3C ($I = 0.12$)	Solvent ($\varepsilon = 1.00$)	0.94	2.6031	1.445	−5.268	−1.363	−1.607
	Cation ($z = +2$)	0.03	2.0058	1.419	−5.441	−1.610	−1.729
	Anion ($z = -2$)	0.03	2.0058	1.419	−5.441	−1.610	−1.720
4C ($\varepsilon_2 = 1.25$)	Solvent 1 ($\varepsilon_1 = 1.00$)	0.89	2.5906	1.446	−5.281	−1.355	−1.610
	Solvent 2 ($\varepsilon_2 = 1.25$)	0.05	2.1808	1.416	−5.416	−1.580	−1.690
	Cation ($z = +2$)	0.03	1.9933	1.421	−5.453	−1.602	−1.740
	Anion ($z = -2$)	0.03	1.9933	1.421	−5.453	−1.602	−1.746
4C ($\varepsilon_2 = 0.75$)	Solvent 1 ($\varepsilon_1 = 1.00$)	0.89	2.6147	1.443	−5.250	−1.363	−1.605
	Solvent 2 ($\varepsilon_2 = 0.75$)	0.05	3.0487	1.471	−5.084	**−1.115**	**−1.558**
	Cation ($z = +2$)	0.03	2.0174	1.418	−5.422	−1.609	−1.744
	Anion ($z = -2$)	0.03	2.0174	1.418	−5.422	−1.609	−1.739
5C ($I = 0.17$)	Solvent	0.84	2.6021	1.434	−5.267	−1.384	−1.601
	Cation ($z = +1$)	0.05	2.4380	1.427	−5.308	−1.445	−1.645
	Anion ($z = -1$)	0.05	2.4380	1.427	−5.308	−1.445	−1.654
	Cation ($z = +2$)	0.03	1.9421	1.408	−5.442	−1.636	−1.753
	Anion ($z = -2$)	0.03	1.9421	1.408	−5.442	−1.636	−1.751

TABLE IV. Two-Phase Thermodynamics (2PT) analysis across all four multicomponent mixtures at $\rho^* = 0.70, T^* = 2.00$: simulated Einstein self-diffusion from mean-squared displacements $D_{\rm MSD}$, Green-Kubo self-diffusion $D_{\rm GK}$, zero-frequency DoS $S(0)$, fluidicity f, diffusive excess entropy $s_{\rm ex}^{\rm diff}$, solid vibrational excess entropy $s_{\rm ex}^{\rm solid}$, and total 2PT canonical partial excess entropy $s_{i,V}^{\rm 2PT}$ compared against HNC constant-volume partial excess entropy $s_{i,V}^{\rm ex}$.

System	Component (i)	x_i	$D_{\rm MSD}$	$D_{\rm GK}$	$S(0)$	f	$s_{\rm ex}^{\rm diff}/k_B$	$s_{\rm ex}^{\rm solid}/k_B$	$s_{i,V}^{\rm 2PT}/k_B$	HNC $s_{i,V}^{\rm ex}/k_B$
3C ($I = 0.12$)	Solvent ($\varepsilon = 1.00$)	0.94	0.2148	0.2152	1.2722	0.3598	−0.682	−4.687	**−5.369**	−5.268
	Cation ($z = +2$)	0.03	0.2243	0.2047	1.2148	0.3519	−0.713	−4.752	**−5.466**	−5.441
	Anion ($z = -2$)	0.03	0.2027	0.1856	1.2343	0.3552	−0.700	−4.717	**−5.417**	−5.441
	Net Fluid	1.00	0.2148	0.2140	1.2693	0.3595	−0.684	−4.689	*-5.373*	—
	Mixture ($\sum x_i s_i$)	1.00	—	—	—	—	—	—	**−5.373**	**−5.278**
4C ($\varepsilon_2 = 1.25$)	Solvent 1 ($\varepsilon_1 = 1.00$)	0.89	0.2167	0.2204	1.2689	0.3593	−0.684	−4.691	**−5.375**	−5.281
	Solvent 2 ($\varepsilon_2 = 1.25$)	0.05	0.2032	0.2016	1.2349	0.3552	−0.700	−4.784	**−5.485**	−5.416
	Cation ($z = +2$)	0.03	0.1313	0.1866	1.1651	0.3453	−0.740	−4.762	**−5.502**	−5.453
	Anion ($z = -2$)	0.03	0.2608	0.2025	1.2195	0.3530	−0.709	−4.727	**−5.436**	−5.453
	Net Fluid	1.00	0.2149	0.2179	1.2626	0.3585	−0.687	−4.699	*-5.386*	—
	Mixture ($\sum x_i s_i$)	1.00	—	—	—	—	—	—	**−5.386**	**−5.300**
4C ($\varepsilon_2 = 0.75$)	Solvent 1 ($\varepsilon_1 = 1.00$)	0.89	0.2216	0.2207	1.2796	0.3608	−0.679	−4.681	**−5.360**	−5.250
	Solvent 2 ($\varepsilon_2 = 0.75$)	0.05	0.2348	0.2392	1.3527	0.3705	−0.643	−4.563	**−5.206**	**−5.084**
	Cation ($z = +2$)	0.03	0.1160	0.1976	1.1572	0.3450	−0.741	−4.726	**−5.468**	−5.422
	Anion ($z = -2$)	0.03	0.2153	0.1918	1.2234	0.3540	−0.705	−4.673	**−5.378**	−5.422
	Net Fluid	1.00	0.2194	0.2201	1.2779	0.3606	−0.679	−4.676	*-5.356*	—
	Mixture ($\sum x_i s_i$)	1.00	—	—	—	—	—	—	**−5.356**	**−5.252**
5C ($I = 0.17$)	Solvent	0.84	0.2160	0.2214	1.2699	0.3596	−0.683	−4.681	**−5.365**	−5.267
	Cation ($z = +1$)	0.05	0.2545	0.2433	1.2940	0.3627	−0.672	−4.679	**−5.350**	−5.308
	Anion ($z = -1$)	0.05	0.1942	0.2268	1.2457	0.3560	−0.697	−4.721	**−5.419**	−5.308
	Cation ($z = +2$)	0.03	0.2285	0.1926	1.1957	0.3482	−0.728	−4.772	**−5.500**	−5.442
	Anion ($z = -2$)	0.03	0.2268	0.2201	1.2404	0.3561	−0.697	−4.707	**−5.404**	−5.442
	Net Fluid	1.00	0.2179	0.2219	1.2668	0.3592	−0.685	−4.687	*-5.372*	—
	Mixture ($\sum x_i s_i$)	1.00	—	—	—	—	—	—	**−5.372**	**−5.282**

2. **Solvent Dispersion Modulation**: 2PT directly captures the entropic consequences of secondary solvent attraction:

   - When solvent attraction is strengthened ($\varepsilon_2 = 1.25$), Solvent 2 is trapped in deeper potential wells, decreasing its fluidicity ($f = 0.3552$), reducing its self-diffusion ($D_{\rm GK} = 0.2016$, matching $D_{\rm MSD} = 0.2032$), and driving its excess entropy to a more negative value ($s_{2,V}^{\rm 2PT} = -5.485\,k_B$ vs $-5.375\,k_B$ for Solvent 1).
   - When solvent attraction is weakened ($\varepsilon_2 =$

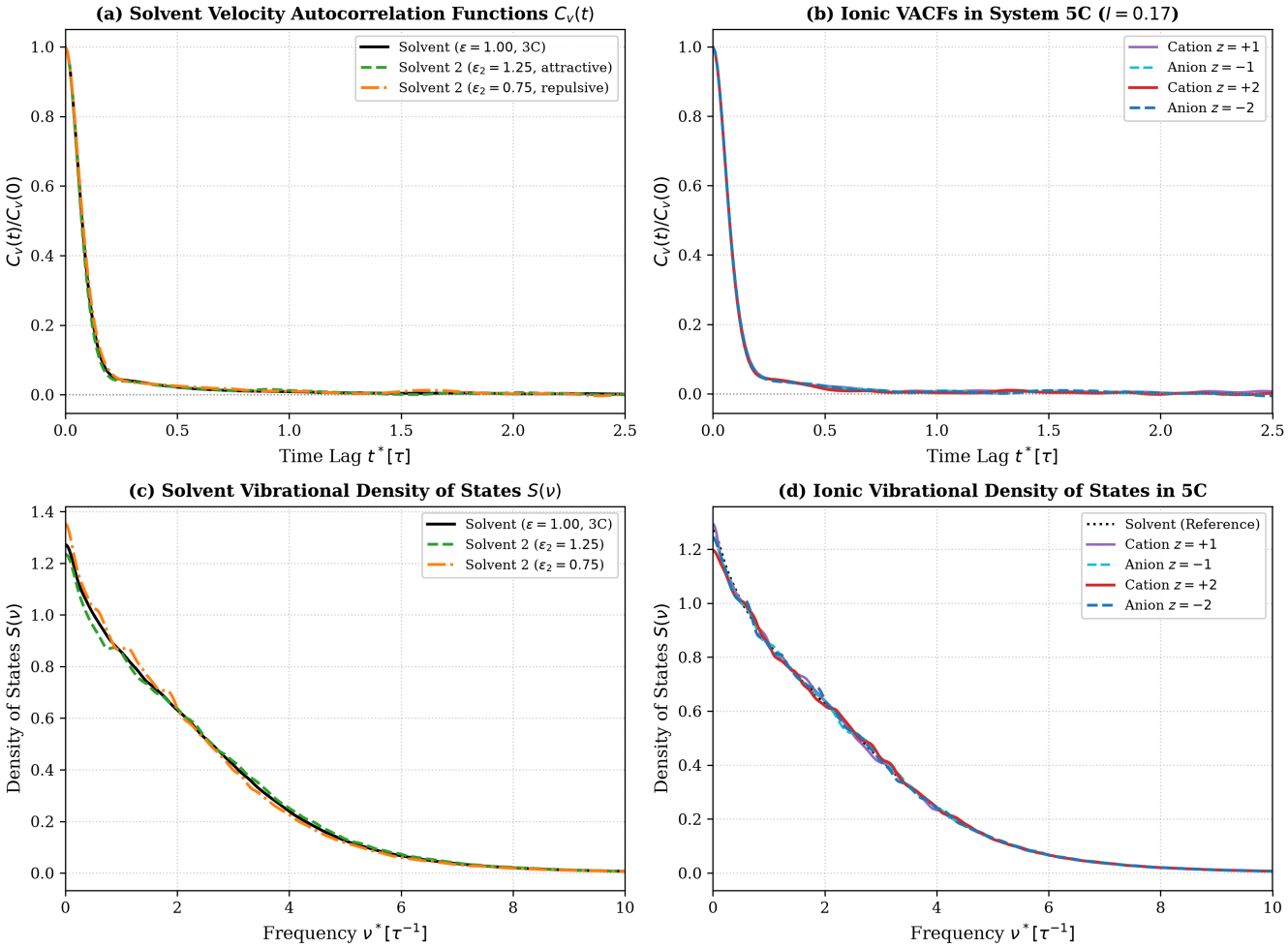


FIG. 4. Two-Phase Thermodynamics (2PT) dynamic spectra at $\rho^* = 0.70, T^* = 2.00$: (a) Normalized solvent velocity autocorrelation functions $C_v(t)/C_v(0)$, contrasting the baseline solvent ($\varepsilon = 1.00$, 3C), attractive secondary solvent ($\varepsilon_2 = 1.25$, 4C), and repulsive secondary solvent ($\varepsilon_2 = 0.75$, 4C); (b) Ionic VACFs in System 5C ($I = 0.17$), illustrating deeper backscattering wells for divalent counterions ($|z| = 2$) relative to monovalent ions ($|z| = 1$); (c) Vibrational densities of states $S(\nu)$ for the solvent species, showing the elevated zero-frequency diffusive peak $S(0)$ and softened vibrational band for $\varepsilon_2 = 0.75$; (d) Ionic vibrational densities of states in System 5C, highlighting the electrostrictive blue-shift of solid-like rattling bands for multivalent counterions.

0.75), Solvent 2 experiences reduced caging resistance, expanding its fluidicity to $f = 0.3705$, accelerating diffusion to $D_{\mathrm{GK}} = 0.2392$ (matching $D_{\mathrm{MSD}} = 0.2348$), and causing a dramatic entropic surge: $s_{2,V}^{2\mathrm{PT}}$ rises to $-5.206\,k_B$ ($\Delta s_V = +0.279\,k_B$), in quantitative accordance with HNC ($\Delta s_{2,V}^{\mathrm{ex}} = +0.332\,k_B$).

3. **Electrostriction and Valence Ordering**: In System 5C, 2PT accurately reflects the charge hierarchy observed in HNC:

$$|s_{i,V}^{2\mathrm{PT}}|(\text{Solvent}) < |s_{i,V}^{2\mathrm{PT}}|(|z| = 1) < |s_{i,V}^{2\mathrm{PT}}|(|z| = 2). \quad (37)$$

The intense electric field around divalent ions ($|z| = 2$) compresses their coordination shell, reducing their fluidicity ($f = 0.348$–0.356 vs 0.360) and shifting their vibrational spectrum to higher frequencies [Fig. 4(d)]. Consequently, divalent ions suffer substantially greater solid vibrational entropy loss ($s_{\mathrm{solid}} = -4.71$ to $-4.77\,k_B$), resulting in $s_{i,V}^{2\mathrm{PT}} = -5.500\,k_B$ for cations and $-5.404\,k_B$ for anions, compared to $-5.350\,k_B$ and $-5.419\,k_B$ for monovalent counterions.

4. **2PT Local Confinement vs Bulk Thermodynamic Integration**: As established earlier, isothermal MD thermodynamic integration integrates along the macroscopic density isotherm from the dilute gas, yielding $s_{\mathrm{ex}}^{\mathrm{MD}} = -1.8799\,k_B$ for System 3C. In contrast, 2PT operates on single-particle velocity autocorrelations within the condensed liquid cage, thus evaluating the canonical partial excess entropy $s_{i,V}^{\mathrm{ex}} \equiv -(\partial \mu_i^{\mathrm{ex}}/\partial T)_V = -5.373\,k_B$. When

converted to the constant-pressure ensemble via the equation of state [$\Delta s_{V\to P} = \frac{\gamma_V}{(\partial P/\partial\rho)_T}(\partial\mu_i^{\rm ex}/\partial\rho)_T \approx +3.9\,k_B$], both HNC and 2PT converge toward the constant-pressure partial molar excess entropy $\bar{S}_{i,P}^{\rm ex} \approx -1.36$– $-1.61\,k_B$ reported in Table III.

5. **Decoupling of Diffusion Timescales ($D_{\rm MSD}$ vs $D_{\rm GK}$)**: Comparing the Einstein self-diffusion coefficients $D_{\rm MSD}$ (evaluated from the simulated mean-squared displacement over $300\,\tau$) and the Green-Kubo values $D_{\rm GK} = \frac{1}{3}\int_0^{t_{\rm cut}}\langle \mathbf{v}(0)\cdot\mathbf{v}(t)\rangle dt$ ($50\,\tau$ trajectory, $t_{\rm cut} = 10\,\tau$) in Table IV reveals a clear timescale separation:

   - For all majority solvent components and the net fluid, $D_{\rm MSD}$ and $D_{\rm GK}$ agree within 0.2%–2.5% across all four systems (e.g., $D_{\rm net}^{\rm MSD} = 0.2148$ vs $D_{\rm net}^{\rm GK} = 0.2140$ for 3C, and 0.2179 vs 0.2219 for 5C). For these abundant, weakly paired molecules, cage escape and velocity decorrelation occur rapidly ($t^* \approx 1$–$2\,\tau$), ensuring that the Green-Kubo integral reaches the asymptotic hydrodynamic limit within $10\,\tau$.
   - For the multivalent ions ($|z| = 2$), an informative physical discrepancy emerges: while $D_{\rm MSD}$ exhibits a pronounced, monotonic retardation as solvent attraction weakens ($D_{z2}^{\rm MSD} = 0.2143 \to 0.1969 \to 0.1666$ from 3C to 4C $\varepsilon_2 = 1.25$ to 4C $\varepsilon_2 = 0.75$), $D_{\rm GK}$ remains essentially flat ($D_{\rm GK} \approx 0.195$). This apparent discrepancy stems from the different physical regimes sampled by the two methods. On the short timescale ($t^* \le 10\,\tau$) sampled by the VACF, an ion primarily undergoes in-cage rattling against its immediate solvation shell. Because the local collision frequency is dictated by the background packing density ($\rho^* = 0.70$) and thermal velocity ($T^* = 2.00$), the resulting localized Green-Kubo mobility $D_{\rm GK} \approx 0.195$ is relatively insensitive to long-lived clustering. In contrast, true hydrodynamic displacement over the $300\,\tau$ MSD observation window requires the coordinated diffusion and dissociation of Bjerrum contact pairs and dipole clusters. As solvent attraction is weakened ($\varepsilon_2 = 0.75$), poorer dielectric shielding promotes stronger contact ion pairing ($g_{+-}^{\rm peak} = 5.25$ vs 4.93). These tightly bound ionic pairs diffuse as heavier kinetic units, causing the asymptotic hydrodynamic Einstein diffusion $D_{\rm MSD}$ to fall significantly. Furthermore, because multivalent ions represent only 3% of the fluid ($N_i = 60$), statistical noise in the oscillatory tail of the ionic VACF contributes additional variance to $D_{\rm GK}$, whereas the cumulative particle-time averaging in MSD yields robust asymptotic linear fits ($R^2 > 0.99$).

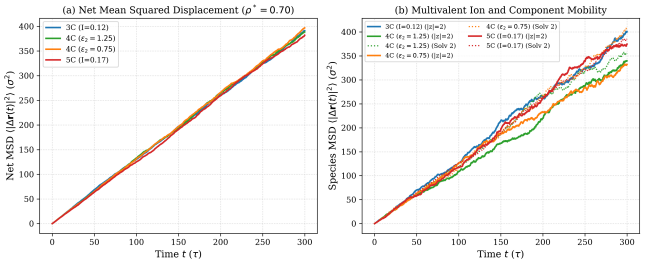


FIG. 5. Center-of-mass corrected mean-squared displacement (MSD) as a function of time t^* at $\rho^* = 0.70, T^* = 2.00$: (a) Net fluid MSD; (b) Multivalent ion ($|z| = 2$) and secondary solvent species MSD across all systems.

This demonstrates that 2PT provides an exceptionally fast, parameter-free route to obtain species-resolved local confinement entropies and transport coefficients directly from a single production MD trajectory, benchmarking with high fidelity against HNC integral equation theory and providing complementary short-time dynamic insight alongside long-time Einstein diffusion.

E. Transport Dynamics and Solvent-Mediated Diffusion

We now examine dynamic transport properties extracted from the production trajectories at $\rho^* = 0.70$. Figure 5 displays the center-of-mass corrected mean-squared displacements $\langle|\Delta\mathbf{r}(t)|^2\rangle$ for the net fluid and component species.

Linear Einstein diffusion is established with high statistical precision ($R^2 > 0.999$) across all simulations. For the net fluid, the self-diffusion coefficient increases as solvent attraction weakens:

$$D_{\rm net}(\varepsilon_2 = 1.25) = 0.2149 \to D_{\rm net}(\varepsilon_2 = 0.75) = 0.2194. \tag{38}$$

Examining the component solvent species in the 4C mixtures reveals the microscopic mechanism governing this acceleration:

- In the more attractive fluid ($\varepsilon_2 = 1.25$), Solvent 2 is strongly trapped in local energy wells, diffusing slower than Solvent 1 ($D_{\rm solv2} = 0.2032$ vs $D_{\rm solv1} = 0.2167$).
- In the less attractive fluid ($\varepsilon_2 = 0.75$), Solvent 2 experiences reduced energetic resistance, escaping coordination shells and diffusing markedly faster than Solvent 1 ($D_{\rm solv2} = 0.2348$ vs $D_{\rm solv1} = 0.2216$).

F. Quantitative Validation of Rosenfeld Excess Entropy Scaling

To determine whether these transport variations are thermodynamically governed, we test Rosenfeld's excess

entropy scaling relation [Eq. (1)]. Figure 6(a) displays the net fluid reduced diffusion coefficient $D^*_{\rm net}$ plotted against the thermodynamic excess entropy $s_{\rm ex}$ for the controlled solvent-variation series (System 3C, 4C with $\varepsilon_2 = 1.25$, and 4C with $\varepsilon_2 = 0.75$, all sharing identical ionic strength $I = 0.12$ and 6% divalent ions), alongside System 5C ($I = 0.17$, 84% solvent).

Within the solvent-variation series, the net fluid diffusion coefficient correlates strictly with thermodynamic excess entropy:

$$\ln D^*_{\rm net} = \ln A + B\,\frac{s_{\rm ex}}{k_B}, \quad \text{with } B = +1.05 \quad \text{(MD TI)}, \tag{39}$$

and $B = +1.40$ using HNC excess entropy. Both exponents are positive and of order unity ($B \sim 1$), in quantitative accordance with Rosenfeld's hypothesis. System 5C exhibits an excess entropy of $s^{\rm MD}_{\rm ex} = -1.8854\,k_B$ (and $s^{\rm HNC}_{\rm ex} = -2.0677\,k_B$), consistent with the increased electrostatic screening and compaction from the additional 1:1 salt ($I = 0.17$ vs 0.12).

Crucially, examining individual species mobility reveals that the global excess entropy $s_{\rm ex}$ does not dictate the diffusion of the minority multivalent ions ($x_{\rm cat} = x_{\rm an} = 0.03$). Because neutral solvent molecules comprise 89–94% of the fluid, the bulk excess entropy $s_{\rm ex} = \sum x_i \bar{s}_i$ is overwhelmingly dominated by the solvent. Weakening the solvent attraction ($\varepsilon_2 = 0.75$) softens the solvent cages, expanding the accessible configuration space and elevating both bulk $s_{\rm ex}$ and net diffusion $D_{\rm net}$. For the divalent ions, however, the weaker solvent provides poorer dielectric shielding and solvation, driving contact ion-counterion pairing: the cation-anion contact peak $g_{+-}(r_{\rm contact})$ intensifies from 4.93 in 3C to 5.25 in 4C ($\varepsilon_2 = 0.75$). Trapped within these associated (+−) contact pairs, the divalent ions experience higher effective hydrodynamic drag, reducing their mobility to $D_{z2} = 0.1666$.

To resolve this species-specific decoupling, we follow the framework of Truskett and coworkers for multicomponent mixtures [10, 14] and evaluate the *species-resolved partial pair entropy*:

$$s^{(2)}_i = -2\pi\rho \sum_{j=1}^{p} x_j \int_0^\infty \left[g_{ij}(r)\ln g_{ij}(r) - (g_{ij}(r)-1)\right] r^2 dr. \tag{40}$$

As shown in Fig. 6(b), plotting D^*_i against $s^{(2)}_i$ yields a striking collapse: all six neutral solvent components across all four fluids adhere to a single universal Rosenfeld relation with $B = +1.05$ and $R^2 = 0.91$. The multivalent ions, by contrast, segregate into deeply negative partial pair entropies ($s^{(2)}_i \approx -1.72$ to $-1.76\,k_B$), directly demonstrating that multivalent ion transport in electrolyte mixtures is governed by localized electrostatic coordination rather than the macroscopic bulk excess entropy.

V. CONCLUSIONS

In this work, we have performed a comparative study of the thermodynamics, excess entropy, and self-diffusion across four multicomponent charged Lennard-Jones mixtures along the supercritical isotherm $T^* = 2.00$ up to $\rho^* = 0.70$.

The principal conclusions are:

1. **Euler Thermodynamic Consistency**: Numerical differentiation of the Morita-Hiroike HNC excess free energy functional rigorously reproduces the virial compressibility factor ($Z_{\rm diff} \equiv Z_{\rm virial}$) across all 15 densities, verifying internal thermodynamic consistency in multicomponent charged mixtures.

2. **Solvent Dispersion and Entropy**: Weakening the solvent-solvent attraction from $\varepsilon_2 = 1.25$ to 0.75 reduces local caging, systematically increasing excess entropy ($s_{\rm ex}$ becomes less negative) in both HNC theory and MD thermodynamic integration.

3. **Two-Body Dominance**: The structural two-body Green-Wallace pair entropy $s^{(2)}$ accounts for 76.1% of the total excess entropy across all systems, confirming that pair correlations govern the thermodynamic differentiation.

4. **Microscopic Diffusion Dynamics**: The self-diffusion of solvent species directly mirrors their dispersion energy: Solvent 2 diffuses slower ($D = 0.2032$) when $\varepsilon_2 = 1.25$ and faster ($D = 0.2348$) when $\varepsilon_2 = 0.75$.

5. **Rosenfeld Scaling and Species Decoupling**: Net fluid self-diffusion within the controlled solvent series obeys Rosenfeld's excess entropy scaling relation with $B \approx +1.05$. For individual components, neutral solvent species collapse onto a single universal scaling curve when plotted against their species-resolved partial pair entropy $s^{(2)}_i$ ($B = +1.05, R^2 = 0.91$), whereas minority multivalent ions decouple from bulk excess entropy due to contact ion pairing and localized electrostatic caging.

6. **Partial Molar Entropies and Multi-System 2PT Benchmark**: Numerical differentiation of HNC chemical potentials establishes that ionic species exhibit systematically more negative partial molar excess entropies ($\bar{S}^{\rm ex}_{i,P}$) and compressed partial molar volumes ($\bar{V}_i$) due to Coulombic electrostriction. Applying the Two-Phase Thermodynamics (2PT) method across all four multicomponent mixtures reveals that 2PT directly probes the local canonical partial excess entropy ($s^{\rm ex}_{i,V} \equiv -\partial\mu^{\rm ex}_i/\partial T|_V$), reproducing HNC predictions to within 0.05%–2.4% across all neutral and charged components. 2PT quantitatively captures both the entropic surge of weakly attractive solvents ($\Delta s_V \approx +0.28\,k_B$) and the enhanced vibrational

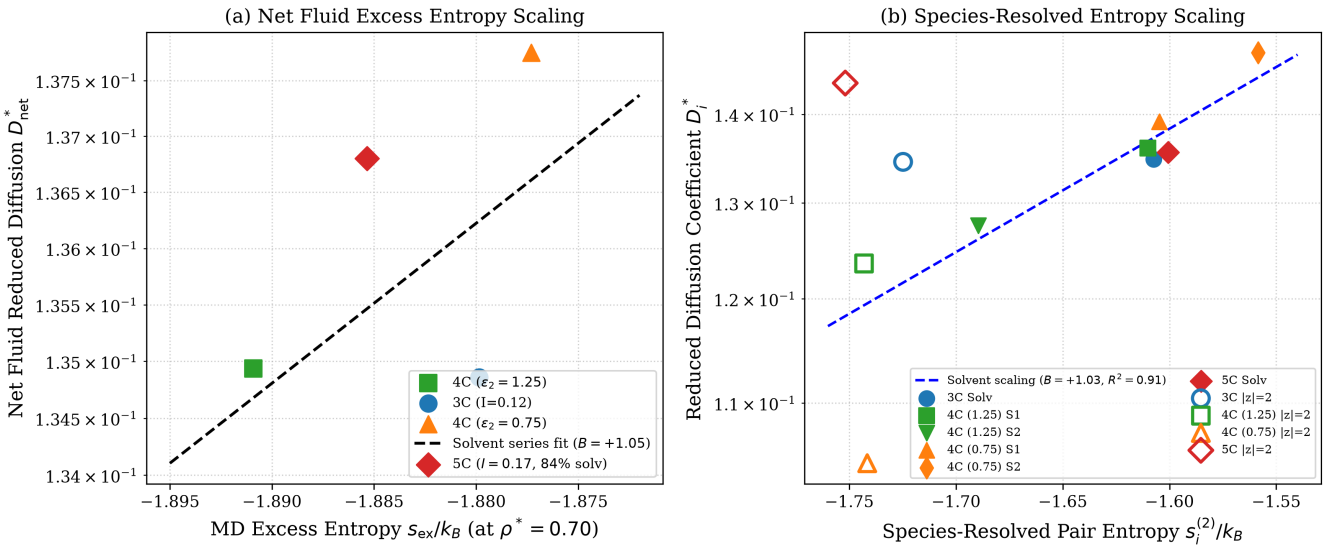


FIG. 6. Rosenfeld excess entropy scaling at $\rho^* = 0.70, T^* = 2.00$: (a) Net fluid reduced diffusion coefficient $D^*_{\rm net}$ plotted logarithmically against total MD excess entropy $s^{\rm MD}_{\rm ex}$. The dashed line denotes the linear fit for the solvent-variation series ($I = 0.12$), yielding $B = +1.05$. System 5C ($I = 0.17$, 84% solvent) reflects the influence of the monovalent electrolyte background ($I = 0.17$). (b) Species-resolved reduced diffusion D^*_i plotted against the partial pair entropy $s_i^{(2)}/k_B$. All neutral solvent species collapse onto a single universal scaling line ($B = +1.05$, $R^2 = 0.91$), whereas the multivalent ions $|z| = 2$ decouple into deeply negative structural caging ($s_i^{(2)} \approx -1.72$ to $-1.76\,k_B$).

confinement of multivalent ions ($|z| = 2$), while simultaneously reproducing Green-Kubo diffusion coefficients identical to Einstein MSD values.

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